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Systems and methods for treating pyronies disease

Abstract

Devices and methods, and for remodeling tissues to treat Peyronie's disease including devices and methods adapted for cryogenic cooling of targeted plaque in a male penis to selectively remodel targeted tissue.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit of priority to U.S. Provisional Patent Application No. 63/136,034 filed Jan. 11, 2021, the content of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to medical devices and methods, and more particularly to cryotherapy devices for remodeling tissues to treat Peyronie's disease. Devices and methods of the invention are adapted for cryogenic cooling of targeted plaque in a male penis to selectively remodel such targeted tissue.

BACKGROUND OF THE INVENTION

(2) Peyronie's disease is a condition resulting from fibrous scar penile tissue that causes curved, painful erections. Peyronie's disease causes a significant bend or pain in some men and can interfere or prevent maintaining an erection leading to erectile dysfunction.

(3) Typically, Peyronie's disease does not go away on its own. In most men having Peyronie's

disease, the condition remains or worsens.

SUMMARY OF THE INVENTION

(4) The techniques described herein relate to methods for cryogenically treating Peyronie's disease, including providing a cryogenic device including at least one tissue-penetrating needle with a cooling tip; advancing the cooling tip into plaque in a subject's penile shaft; and cooling the plaque with the cooling tip thereby inducing a cooling injury to the plaque to achieve a therapeutic effect.

(5) The techniques described herein can relate to a method wherein cooling the plaque with the cooling tip maintains the cooling tip at a targeted cooling temperature for at least 5 seconds. This targeted cooling temperature can be lower than -10 degrees Celsius and, in some variations, is between -10 and -80 degrees Celsius.

(6) In additional variations, the techniques described herein relate to a method wherein cooling the plaque with the cooling tip includes maintaining the cooling tip in a stationary position in the plaque. In additional variations, the therapeutic treatment allows the plaque to be absorbed by a body of a patient.

(7) The treatments described herein can result in reduced penile curvature.

(8) Variations of the method can include imaging the plaque to provide a 3-dimensional map of the plaque. For example, such imaging can include the use of an ultrasound device. In additional aspects, the techniques and methods described herein relate to imaging during advancing the cooling tip.

(9) In some aspects, the techniques described herein relate to a method further including a controller operatively coupled to the cryogenic device and the ultrasound device. Additional variations include methods wherein the controller is configured to modulate cooling the plaque in response to imaging signals from the ultrasound device.

(10) In additional variations, the methods and techniques described herein include a controller that is configured to (i) record a 3-dimensional map of the plaque provided by the ultrasound device, (ii) monitor location of one or more cooling tips within the plaque with the ultrasound device; and (iii) actuate the cryogenic device to cool the plaque; and (iv) terminate cooling with the cryogenic device in response imaging with the ultrasound device.

(11) Variations of the systems and methods can include a warming system with at least one warming needle with a warming tip and introducing a warming tip into tissue adjacent to the plaque. In some aspects, the methods can further include warming an adjacent tissue with the warming tip to thereby prevent a cooling injury to the adjacent tissue.

(12) The warming can be performed prior to cooling the plaque, contemporaneous with cooling the plaque, and/or after cooling the plaque.

(13) Additionally, the warming tip can be operatively coupled to the controller, and the controller is configured to modulate warming the adjacent tissue. In addition, or as an alternative, the method can include modulating warming of the adjacent tissue in response to signals from a temperature sensor carried by the warming tip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic elevational view of a cryogenic probe and system corresponding to the invention with a needle assembly detachably coupled to a handpiece component together with an optional controller component.

(2) FIG. 2 is an enlarged cut-away schematic view of the cryogenic probe of FIG. 1, showing a cooling fluid injector tube in relation to the lumen in the needle.

(3) FIG. 3 is an illustration of a variation of the cryogenic probe of FIG. 1 with a needle assembly consisting of a plurality of tissue-penetrating needles.

(4) FIG. 4 illustrates a penile shaft of a subject with Peyronie's disease wherein the needle of the probe of FIG. 1 is shown after penetration into the plaque for providing cryogenic treatment together with ultrasound apparatus positioned to image the formation of an ice-ball in the plaque.

(5) FIG. 5 is another view of a penile shaft with Peyronie's disease a probe with a plurality of cryogenic needles introduced into plaque in a penile shaft.

DETAILED DESCRIPTION OF THE INVENTION

(6) Referring now to FIG. 1, a cryogenic treatment system **100** for cryogenic treatment of tissue is shown, which comprises a cryogenic probe **105** with a proximal handle or handpiece **106** coupled to a tissue-penetrating cryogenic needle **110**. The handpiece **106** is adapted for gripping by human hand or can be configured for attachment to a robotic assembly as is known in the art. The cryogenic system optionally includes a controller **112** coupled to the probe **105** as will be described further below.

(7) As shown in FIG. 1, a cryogenic cooling fluid assembly is carried within the handpiece **106**, which typically consists of a single-use cooling cartridge **115** with and valve mechanism **116** operatively coupled to an electrical power source **120** adapted to open and close the valve mechanism **116**. The electrical power source **120** further is coupled to a processor or subcontroller **125** for controlling flows of a cooling fluid **126** from the cartridge **115** to the tissue-penetrating needle **110**. In one variation, the subcontroller **125** may be carried on a single processor board in the handpiece **106** and is adapted to perform one or more selected programs. The subcontroller **125** can comprise a programmable microprocessor that carries computer code or programming instructions for a treatment cycle, wherein such a treatment cycle typically comprises an on/off interval which delivers the cooling fluid **126** at a predetermined flow rate for a predetermined time interval which then provides a treatment of a predetermined volume a tissue which comprises the creation of an ice-ball in the targeted tissue or plaque as will be described below.

(8) The power source **120** and valve mechanism **116** are typically activated manually by a switch **128** in the handpiece **106** that triggers the controller **125** to control a treatment cycle. The power source **120** can comprise a rechargeable battery or single-use battery that actuates, for example, a solenoid-type of valve mechanism **116**.

(9) Referring to FIGS. 1 and 2, in one variation, the cryogenic probe **105** carries a single hollow, tissue-penetrating needle **110** that may be detachably coupled to the handpiece **106** with coupling **132**. A cooling fluid path **140** is shown in FIG. 1 that extends from cooling fluid source or cartridge **115** to the distal region **142** of the needle **110**.

(10) In a variation, still referring to FIGS. 1 and 2, the needle **110** can consist of a 30-gauge hollow hypotube having a sharpened, closed-end distal tip **145**. The needle **110** can have any suitable axial length between the handpiece **106** and the distal tip **145** of the needle **110** ranging from 5 mm to 50 mm. Typically, the needle has a length from about 5 mm to about 20 mm although any length is possible. Such a needle **110** can be straight or have a curved distal portion. Such a needle **110** can comprise a stainless-steel material with an inner diameter of about 0.005" and an outer diameter of about 0.010" to 0.015". Alternative probes can carry multiple needles having other outer diameters from about 0.006 inches to about 0.100 inches (see FIG. 3). Typically, the needles will be a 16 gauge or smaller.

(11) Referring to FIG. 1, in one variation, the exemplary cooling fluid supply or cartridge **115** contains a liquid cooling fluid **126** under high pressure, wherein the liquid preferably has a boiling temperature that is lower than body temperature (37° C.). Thus, when the cooling fluid **126** is delivered through the tissue-penetrating needle **110** when penetrated into targeted tissue, the heat from the targeted tissue will evaporate the liquid cooling fluid **126** within the needle **110**, resulting in cooling the target tissue typically with the formation of an ice-ball **150** in the targeted tissue or plaque **155** (see FIG. 4). The valve **116** is provided within the handpiece **106** in the cooling fluid flow pathway **140** between the cartridge **115** and needle tip **145**. Typically, the subcontroller **125** is configured to limit the cooling fluid flow rate and cooling fluid volume in a treatment cycle, which

in turn controls the rate of temperature change the targeted tissue, and thereby controls the dimensions of the ice-ball **150** that is formed in the targeted tissue or plaque **155** (FIG. **4**). The subcontroller **125** thus controls the pressure of the cooling fluid **126** delivered into the needle **110** and the temperature of the needle tip in contact the targeted tissue thus can be controlled. A mechanical pressure relief valve **156** also may be used to control the pressure within the lumen **158** of the needle **110** (FIG. **2**).

(12) In FIG. **1**, the cooling fluid **126** is carried in a single-use cartridge **115** with a metal cap or seal, and in one variation, the cartridge contains liquid N₂O. Other cooling fluids also can be used, where exemplary cooling fluids include fluorocarbon refrigerants and/or carbon dioxide. The volume of cooling fluid **126** contained by cartridge **115** typically is adequate to treat plaque **155** in a Peyronie's disease patient in a single procedure. An exemplary liquid N₂O cartridge can contain, for example, a cooling fluid volume in a range of 10 grams to 100 grams of a cooling liquid.

(13) Referring now to FIG. **2**, the flow of cryogenic cooling fluid **126** from the cartridge **115** is controlled by the valve **116**, which can comprise an electrically actuated solenoid valve or the like operating in response to control signals from the subcontroller **125** (FIG. **1**). A typical valve **116** can be adapted for on/off operation and may provide for venting of the cooling fluid path downstream from the valve **116** after the valve is closed, which can limit residual cryogenic fluid vaporization and cooling.

(14) In FIG. **2**, the cryogenic cooling fluid **126** is released via valve **116** to flow through an injection tube **165** that communicates with the cooling fluid path **140** in the handpiece **106**. The injection tube **165** is carried within the lumen **166** of the needle **110** and wherein the distal end **168** of the injection tube **165** extends close to the distal end **170** of the lumen **166** in the needle **110**. The injection tube **165** can comprise a metal or polymer material or a combination thereof. The injection tube **165** has an outer diameter that is less than the diameter of the lumen **166** in the needle **110** such that outflows of a cooling fluid **126** (at least partly comprising a gas in the outflow) will be accommodated in the annular space **172** between the injection tube **165** and the wall of the needle **110**. As an example, the injection tube **165** can have an inner lumen diameter ranging between 10 μ m and 100 μ m. An outer diameter of the injection tube **165** will typically be less than about 1000 and often being less than about 500 μ m.

(15) Referring now to FIGS. **2** and **3**, it can be understood that the cooling fluid **126** is injected into lumen **166** of needle **110** and as the liquid cooling fluid vaporizes within the needle **110**, such vaporization will cool and freeze the tissue or plaque **155** in contact with the needle **110**.

(16) Referring now to FIG. **3**, an alternative embodiment of a cryogenic system **100'** and cryogenic probe **105'** is shown that is similar to that of FIG. **1** except that the fluid path **140'** from the cooling fluid cartridge **115** communicates with a plurality of needles and in this variation consist of two spaced-apart tissue-penetrating needles **175a** and **175b**. It should be appreciated that the number of such needles can range from 1 to 6 and operate as described above. In all other respects, the variation of FIG. **3** operates as described previously.

(17) Referring back to FIGS. **1** and **2**, the cartridge **115** can be initially inserted into the handpiece **106** and be adapted for use by piecing a metal cap of the cartridge **115** as is known in the art. Further, one or more filters (not shown) can be provided in the fluid path **140**.

(18) It should also be appreciated that the proximal portion **176** of the needle **110** (FIG. **1**) can be supported and surrounded by an insulator member **177** adapted to limit heat transfer from the proximal portion **176** of the needle **110** to the environment. In other variation, the needle or needles can have flat or oval cross-sectional shapes which can be desirable for creating suitable ice-balls in tissue.

(19) Referring now to FIG. **4**, a method of treating Peyronie's disease is shown using the probe **105** of FIG. **1** where the objective is to cryogenically treat plaque **155** in a penile shaft **182** of the patient. In one method variation, the needle **110** is penetrated into the plaque **155** in an orientation that is generally perpendicular to the axis **185** of the penile shaft **182**. In this variation, first and

second ultrasound transducers **188A** and **188B** (also shown in FIG. 1) are positioned approximately 90° apart from another to provide for bi-planar views of the plaque **155** to optimally position the needle **110** in the plaque **155**.

(20) The ultrasound transducers **188A** and **188B** are subsequently used for observing the formation of the ice ball **150** in the plaque **155**. During such a procedure, the physician would penetrate the needle **110** into the plaque **155** sequentially in multiple locations under ultrasonic monitoring and actuate the subcontroller **125** (FIG. 1) to deliver a predetermined cooling dose, which would be selected based on evaluation of the dimensions of the plaque prior to insertion of the needle **110**. Following treatment, the patient would optionally tension the penile shaft **182** in a straightened position with traction devices that are known in the art. Within about two to six weeks following the procedure, the patient's immune system would absorb or resorb the treated plaque **155** and would reduce the volume of the plaque **155** in the range of 70% to 90%. With a reduction in the volume of the plaque **155**, the penile shaft **182** would be straightened. The use of traction on the penile shaft **182** can be used for any suitable time interval post-treatment, for example, one to two weeks.

(21) FIG. 5 illustrates another method of the invention for treating Peyronie's disease wherein the probe **100'** has two needles **190a** and **190b** that are similar to the variation of FIG. 3. In the method shown in FIG. 5, it can be seen that the two needles **190a** and **190b** can be introduced from a lateral side of the plaque **155** rather than directly into the plaque from a superior position as shown in FIG. 4. In this case, dual ultrasound transducers **188A** and **188B** again can be positioned such that the needles **190a** and **190b** can be inserted into the plaque **155** prior to performing a treatment cycle. In this case, the dual needles **190a** and **190b** are adapted to create a more planar ice-ball **150'** for ablating the plaque **155**, which typically may form as a somewhat flat layer in the penile shaft **182**.

(22) In general, a method corresponding to the invention for cryogenically treating Peyronie's disease comprises providing a cryogenic device including at least one tissue-penetrating needle with a cooling tip, advancing the cooling tip into plaque in a subject's penile shaft, and cooling the plaque with the cooling tip thereby inducing a cooling injury to the plaque to achieve a therapeutic effect. Typically, the cooling step of the method maintains the cooling tip at a targeted cooling temperature for at least 5 seconds. The targeted cooling temperature is lower than -10 degrees Celsius and typically the temperature is between -10 degrees Celsius and -80 degrees Celsius.

(23) In one variation, the cooling step of the method includes maintaining the cooling tip **145** in a stationary position in the plaque. Alternatively, the cooling step includes moving the cooling tip **145** in the plaque **155**. Following the cooling step, the therapeutic result is achieved wherein the plaque is absorbed or resorbed by the patient's body and can reduce penile curvature.

(24) In a variation, the method can include imaging the plaque to provide a 3-dimensional map of the plaque, typically with one or more ultrasound devices or transducers such as ultrasound transducers **188A** and **188B** shown in FIGS. 4 and 5. Such an imaging step can be used as the needle(s) and cooling tip **145** are advanced into the plaque **155** and during the cooling step to observe formation of the ice-ball **150** in the plaque **155**.

(25) In another variation of a treatment system and method, the system controller **112** is operatively coupled to the cryogenic probe **105** and the ultrasound devices or transducers **188A** and **188B**. In this variation, the system controller **112** is configured to modulate the cooling step in response to imaging signals from the ultrasound transducers **188A** and **188B**. Typically, the system controller **112** is configured to (i) record a 3-dimensional map of the plaque provided by the ultrasound system, (ii) monitor a location of one or more cooling tips within the plaque with the ultrasound system, (iii) actuate the cryogenic device to cool the plaque, and (iv) terminate cooling with the cryogenic device in response imaging with the ultrasound system.

(26) In another variation, the treatment method further comprises the use of a tissue-warming system **200** with at least one warming needle **210** with a warming tip **215**, wherein the warming tip **215** is introduced into tissue adjacent the plaque **155** to thereby prevent a cooling injury or freezing

of such adjacent tissue. Such a method of warming tissue can be performed (i) prior to the step of cooling the plaque, (ii) contemporaneous with the step of cooling the plaque, and/or (iii) after the step of cooling the plaque. Such a warming system can be operatively coupled to the system controller **112** and such a controller can be configured to modulate the warming step. In one variation, the warming step is modulated in response to signals from a temperature sensor **218** carried by the warming tip **215** of the at least one warming needle **210**.

Claims

1. A method for cryogenically treating Peyronie's disease, comprising: providing a cryogenic device including at least one tissue-penetrating needle with a cooling tip; advancing the cooling tip into plaque in a subject's penile shaft; and cooling the plaque with the cooling tip thereby inducing a cooling injury to the plaque to achieve a therapeutic effect by reducing curvature of the penile shaft.
2. The method of claim 1, wherein cooling the plaque with the cooling tip maintains the cooling tip at a targeted cooling temperature for at least 5 seconds.
3. The method of claim 2, wherein cooling the plaque with the cooling tip comprises cooling to a targeted cooling temperature lower than -10 degrees Celsius.
4. The method of claim 2, wherein the targeted cooling temperature is between -10 and -80 degrees Celsius.
5. The method of claim 1, wherein cooling the plaque with the cooling tip includes maintaining the cooling tip in a stationary position in the plaque.
6. The method of claim 1, wherein cooling the plaque includes moving the cooling tip in the plaque.
7. The method of claim 1, wherein the therapeutic effect comprises the plaque being absorbed by a body of a patient.
8. The method of claim 1, further comprising imaging the plaque to provide a 3-dimensional map of the plaque.
9. The method of claim 8, wherein imaging the plaque is performed with an ultrasound device.
10. The method of claim 8, further comprising imaging during advancing the cooling tip.
11. The method of claim 8, further comprising imaging during cooling the plaque.
12. The method of claim 8, further comprising imaging with an ultrasound device.
13. The method of claim 12, further comprising a controller operatively coupled to the cryogenic device and the ultrasound device.
14. The method of claim 13, wherein the controller is configured to modulate cooling the plaque in response to imaging signals from the ultrasound device.
15. The method of claim 14, wherein the controller is configured to (i) record a 3-dimensional map of the plaque provided by the ultrasound device, (ii) monitor location of one or more cooling tips within the plaque with the ultrasound device; and (iii) actuate the cryogenic device to cool the plaque; and (iv) terminate cooling with the cryogenic device in response imaging with the ultrasound device.
16. The method of claim 15, further comprising a warming system with at least one warming needle with a warming tip and introducing a warming tip into tissue adjacent the plaque.
17. The method of claim 16, further comprising warming an adjacent tissue with the warming tip to thereby prevent a cooling injury to the adjacent tissue.
18. The method of claim 17, wherein warming is performed prior to cooling the plaque, contemporaneous with cooling the plaque, and/or after cooling the plaque.
19. The method of claim 18, wherein the warming tip is operatively coupled to the controller and the controller is configured to modulate warming the adjacent tissue.

20. The method of claim 18, wherein the warming the adjacent tissue is modulated in response to signals from a temperature sensor carried by the warming tip.
